



Drug-Eluting Resorbable Scaffold Versus Balloon Angioplasty for Below-the-Knee Peripheral Artery Disease: 2-Year Results From the LIFE-BTK Trial

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BACKGROUND: Limited treatment options exist for infrapopliteal disease in patients with chronic limb-threatening ischemia (CLTI), a condition associated with a high risk of limb loss. Interventional management of diseased infrapopliteal vessels with percutaneous transluminal angioplasty (PTA) is associated with high rates of restenosis and reintervention. In the LIFE-BTK randomized controlled trial (Pivotal Investigation of Safety and Efficacy of BRS Treatment-Below the Knee), the drug-eluting resorbable scaffold (DRS) demonstrated superior 12-month efficacy compared with PTA in a selected CLTI population with predominantly noncomplex, mildly to moderately calcified lesions. This report presents the 2-year safety and efficacy outcomes of the Esprit BTK DRS system in the LIFE-BTK randomized trial comparing DRS with PTA for treatment of infrapopliteal vessels and CLTI.

METHODS: The LIFE-BTK trial was a multicenter, subject-blinded, randomized controlled trial enrolling 261 patients with CLTI who were randomized 2:1 to receive either DRS or PTA. The revised primary efficacy end point was freedom from target limb amputation, target vessel occlusion, clinically driven target lesion revascularization, or binary restenosis. The primary safety end point was freedom from major adverse limb events and perioperative death. Predictors of efficacy and clinically driven target lesion revascularization were analyzed along with subgroup assessments.

RESULTS: At 2 years, the primary efficacy end point was observed in 68.8% of the DRS group versus 45.4% of the PTA group ($P=0.0004$). Limb salvage rates were 94.7% for DRS and 97.3% for PTA ($P=0.34$). Binary restenosis occurred in 28.5% of DRS patients versus 48.2% of PTA patients ($P=0.005$), and clinically driven target lesion revascularization rates were 9.7% versus 18.6%, respectively ($P=0.034$). The primary safety end point was observed in 91.6% of the DRS group versus 95.6% of the PTA group ($P=0.16$). Scaffold treatment was an independent predictor of efficacy (odds ratio, 0.27; $P=0.0003$) and showed a trend toward reduced risk of clinically driven target lesion revascularization, though this did not reach statistical significance. Other predictors included lesion length, Rutherford-Becker class 5, total occlusion, previous amputation, preintervention stenosis, and number of wounds. Subgroup analyses demonstrated consistent efficacy across various patient populations.

CONCLUSIONS: At 2 years, the Esprit BTK DRS demonstrated improved efficacy compared with PTA in maintaining arterial patency, preventing restenosis, and reducing revascularization rates while maintaining a comparable safety profile. These findings support the Esprit BTK scaffold as a promising treatment option for appropriately selected patients with infrapopliteal artery disease and CLTI.

REGISTRATION: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04227899.

Key Words: balloon angioplasty ■ binary restenosis ■ chronic limb-threatening ischemia ■ DRS ■ drug-eluting resorbable scaffold ■ infrapopliteal artery disease ■ vascular patency

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Clinical Perspective

What Is New?

- This report demonstrates higher composite rates of patency and limb salvage with the Esprit BTK drug-eluting resorbable scaffold (DRS) compared with percutaneous transluminal angioplasty alone in patients with chronic limb-threatening ischemia.
- The safety and efficacy of the Esprit BTK DRS are sustained between 12 and 24 months and consistent across multiple patient subgroups.
- Sustained patency of the Esprit BTK DRS results in fewer reinterventions compared with standard endovascular care.

What Are the Clinical Implications?

- There is growing recognition that balloon angioplasty alone may be insufficient for treating BTK disease, given its high rates of restenosis and reintervention.
- The Esprit BTK DRS is designed to address both mechanical (eg, dissection, recoil, and residual plaque) and biological (eg, intimal hyperplasia) contributors to patency loss in tibial interventions.
- In the LIFE-BTK trial, use of the Esprit BTK DRS was associated with improved wound healing and sustained relief of rest pain with fewer interventions compared with percutaneous transluminal angioplasty.

Patients with chronic limb-threatening ischemia (CLTI) present with impaired perfusion of the lower extremities and clinical signs and symptoms of ischemic rest pain, nonhealing wounds, and gangrene.¹ Avoidance of amputation in these patients typically requires restoration of adequate tissue perfusion, which has historically been achieved through surgical bypass.^{2,3} In recent decades, multiple studies have demonstrated the use of endovascular therapy, which can achieve effective revascularization while minimizing the physiological stress associated with open surgery.^{4–6}

Although endovascular intervention has become an accepted alternative to bypass surgery in many patients, the high prevalence of infrapopliteal disease in CLTI presents unique challenges. These include limited endovascular treatment options and high rates of restenosis with therapies such as percutaneous transluminal angioplasty (PTA).^{3,7} PTA remains the most commonly used modality for infrapopliteal revascularization, performed in approximately 70% of cases in the United States,^{8–10} but its long-term efficacy is hindered by several limitations.

Since the design and US Food and Drug Administration protocol approval of the LIFE-BTK randomized controlled trial (Pivotal Investigation of Safety and Efficacy of BRS Treatment-Below the Knee) in 2018 to 2019, the CLTI treatment landscape has continued to evolve. Notably, the BEST-CLI trial¹¹ provided level 1 evidence

Nonstandard Abbreviations and Acronyms

BTK	below the knee
CD-TLR	clinically driven target lesion revascularization
CEC	clinical events committee
CLTI	chronic limb-threatening ischemia
DAPT	dual antiplatelet therapy
DRS	drug-eluting resorbable scaffold
DS	diameter stenosis
DUS	duplex ultrasound
HR	hazard ratio
KM	Kaplan-Meier
OR	odds ratio
PTA	percutaneous transluminal angioplasty
RBC	Rutherford-Becker class
TASC II	Trans-Atlantic Inter-Society Consensus II

supporting surgical bypass with an autologous conduit in appropriately selected patients, reinforcing the importance of individualized treatment strategies. However, at the time of trial initiation, PTA remained the most widely accepted comparator for endovascular therapies in below-the-knee (BTK) disease.

An effective therapy for infrapopliteal arteries must address the heavy calcific plaque burden, the risk of recoil or dissection, and the development of intimal hyperplasia, all of which are inadequately managed by PTA alone.¹² Drug-eluting resorbable scaffolds (DRSs) offer a novel approach by providing temporary mechanical support, delivering antiproliferative therapy during the restenotic phase, and fully resorbing to allow for vessel restoration and future interventions.¹² In the LIFE-BTK trial,¹³ the Esprit BTK DRS everolimus-eluting resorbable scaffold (DRS) demonstrated superior efficacy compared with PTA at 12 months, leading to US Food and Drug Administration approval and subsequent clinical adoption. In this trial, DRS demonstrated safety comparable to PTA while improving vessel patency and limb salvage at 1 year.

Despite these promising results, long-term limb preservation requires sustained patency of infrapopliteal vessels beyond 1 year, a critical factor for complete wound healing and durable relief of ischemic rest pain symptoms. This report presents the 2-year outcomes from the LIFE-BTK trial, evaluating the safety and efficacy of the Esprit BTK scaffold in patients with CLTI, including vessel patency rate, restenosis prevention, and need for reintervention over the follow-up period.

METHODS

Data Sharing Statement

The data, methods used in this analysis, and materials used to conduct the research will not be made publicly available. The authors

will provide data for reproducing the results for the purposes of audit or governance under a confidentiality agreement. Any relevant inquiry should be emailed to the corresponding author.

Study Design

The LIFE-BTK trial was a multicenter, subject-blinded, randomized controlled trial conducted across 50 sites in 6 countries between July 2020 and September 2022. It evaluated the efficacy and safety of a new everolimus-eluting resorbable scaffold (Esprit BTK DRS) compared with PTA in patients with CLTI and infrapopliteal artery disease. Patients aged 18 years or older with Rutherford-Becker class (RBC) 4 (ischemic rest pain) or 5 (minor tissue loss) disease were eligible if they had infrapopliteal artery stenosis ($\geq 70\%$) or occlusion in the proximal two-thirds of the lower leg with at least 1 runoff vessel to the ankle free of clinically significant disease. Key exclusion criteria included acute limb ischemia, inability to tolerate dual antiplatelet therapy (DAPT), and use of atherectomy or specialty balloons for treatment of the target lesion(s). All eligibility criteria were met before randomization. Participants were randomly assigned at a 2:1 ratio to receive either the DRS or PTA after successful treatment of inflow and nontarget lesions and guidewire crossing of the target lesion. Inflow lesions were defined as those located in the common iliac, external iliac, common femoral, superficial femoral, and popliteal arteries. The randomization sequence was managed via electronic data capture (Oracle Clinical). The study protocol was approved by the institutional review board at each participating site, and all patients provided written informed consent. Further details of the study methodology have been previously published.^{13,14} The trial (LIFE-BTK randomized controlled trial) was registered on ClinicalTrials.gov (registration: URL: <https://www.clinicaltrials.gov>; Unique identifier: NCT04227899) on January 14, 2020.

Study Device and Procedure

The Esprit BTK DRS, designed for infrapopliteal use, consists of a poly(L-lactide) backbone coated with a layer containing everolimus and poly(D,L-lactide) with platinum radiopaque markers at both ends. It is bioresorbable, allowing vessel remodeling over time. In the DRS group, predilatation was mandatory; the use of a noncompliant balloon with a 1:1 ratio of balloon-to-vessel diameter was preferred, achieving residual stenosis $< 30\%$ before scaffold placement. Postdilatation was recommended, with noncompliant balloons sized to remain within the margins of the scaffold. The total scaffolded length was limited to 170 mm per patient, and tandem lesions separated by < 3 cm were considered a single target lesion. Angioplasty was performed using standard-of-care techniques, with balloon selection, inflation pressure, and duration at the discretion of the proceduralist. Intravascular ultrasound or optical coherence tomography was encouraged in both groups for procedural optimization. DAPT was recommended for at least 1 year in the DRS group and for 1 month in the PTA group, followed by single-agent therapy in both groups thereafter.

Follow-Up and End Point Definitions

Scheduled follow-ups occurred at 30 days, 3 months, 6 months, 1 year, and annually thereafter up to 5 years. At 2 years (± 28 days), an in-office visit was performed, including duplex

ultrasound (DUS) of the target vessel, ankle brachial index/toe brachial index measurement, RBC assessment, medication use, adverse events, and wound assessment.

The primary efficacy end point was a composite of primary patency and limb salvage, defined as freedom from target limb amputation above the ankle, total occlusion of the target vessel, clinically driven target lesion revascularization (CD-TLR), or binary restenosis of the target lesion. The original primary efficacy end point did not include freedom from binary restenosis of the target lesion and was initially assessed at 6 months. Before completion of enrollment, the end point was revised to include freedom from binary restenosis, and the observation period was extended to 12 months. This change was made following a prespecified blinded interim analysis and in consideration of end points used in recent randomized trials, which were found to be underpowered to detect clinically meaningful differences between the investigational device and standard of care. Binary restenosis was defined as $> 50\%$ diameter stenosis (DS) by angiography or a peak systolic velocity ratio ≥ 2.0 on DUS. Each target lesion was assessed for a raised peak systolic velocity ratio. The core laboratory then used additional secondary criteria to confirm target lesion stenosis. These criteria included visible stenosis on B-mode imaging, focal increase in absolute peak velocity, poststenotic turbulence, change in waveform shape, or velocity drop distal to the stenosis. If peak systolic velocity ratio could not be calculated, then these secondary factors were used to determine significant stenosis. If indeterminate (discordant), then the subject was excluded from the analysis. If a subject underwent both angiogram and DUS at the same time, then the angiogram was the primary determinant of binary restenosis. Two secondary end points were statistically powered for the 1-year analysis: (1) a composite of freedom from above-ankle amputation, target vessel occlusion, and CD-TLR and (2) binary restenosis of the target lesion. These end points were not powered for the 2-year follow-up period; analyses at 2 years are descriptive and exploratory in nature. The primary safety end point included freedom from major adverse limb events at 2 years and perioperative death at 30 days. A major adverse limb event was defined as above-ankle amputation or major reintervention (new surgical bypass grafting, interposition grafting, thrombectomy, or thrombolysis related to the target lesion). Adverse events were adjudicated by a blinded clinical events committee (CEC). Core laboratories evaluated angiographic, DUS, and wound healing outcomes.

Statistical Analysis

The primary efficacy end point was analyzed in the intention-to-treat population, which included all randomized patients who had either qualified 2-year imaging data (angiography or DUS) adjudicated by the core laboratory or a primary efficacy end point event. The primary safety end point was analyzed in the as-treated population, comprising patients who received the assigned treatment. Event rates were calculated as the percentage of patients free from events through 2 years, with only the first occurrence of an event considered for composite end points. The rate of binary restenosis of the target lesion at 2 years was presented as the event rate, including only the first occurrence of restenosis through 2 years after the index procedure. Kaplan-Meier (KM) analyses were used to estimate event-free survival probabilities for efficacy and safety end points.

Both binary event rates and KM estimates were used to summarize outcomes. Binary rates, calculated as the proportion of patients with events at a fixed time point, are presented for clarity and ease of interpretation, particularly in tabular summaries. KM estimates were used for time-to-event analyses to account for censoring and variable follow-up durations, providing a more accurate estimation of event-free probabilities over time. Differences between treatment groups were assessed using Cox proportional hazards regression models, and hazard ratios (HRs) with corresponding 95% CIs were reported. For binary end points, the Newcombe score method¹⁵ was used to calculate differences and 95% CIs. Patients who left the study before the lower limit of the 2-year follow-up window (702 days) without experiencing a primary efficacy event were excluded from denominator calculations. For the components of the above-ankle amputation in the index limb, major reintervention on the index limb, and CD-TLR, data were based on CEC adjudication through 730 days after the index procedure. Imaging end points such as 100% total occlusion of the target vessel and binary restenosis were based on CEC adjudication or core laboratory imaging data, including unscheduled assessments up to 730 days if determined by event date, 788 days for angiography, and 818 days for DUS. For end point rate calculations, the hierarchy was as follows. CEC adjudication results took precedence. If there was no associated adverse event, then CEC adjudication was not performed, and core lab adjudication was used instead. When both angiogram and DUS imaging data were available for the same time point, angiogram data took precedence over DUS for evaluating the primary efficacy end point and the secondary end points. Multivariable Cox regression models were used to identify predictors of the primary efficacy end point at 2 years. Variables were selected based on univariable significance ($P < 0.20$) and included in a stepwise model-building process, with entry and removal thresholds set at $P = 0.20$ and $P = 0.10$, respectively. The proportional hazards assumption was checked for the treatment arm indicator. A multiplicative interaction model was employed to assess the interaction between the treatment arm and a subgroup within the generalized linear model framework. Missing data were handled with multiple imputation. Statistical significance for powered end points was defined using a 1-sided alpha level of 0.025. Analyses were conducted using SAS software (version 9.4, SAS Institute).

RESULTS

Baseline Patient Characteristics

The study 2:1 randomized 261 patients with CLTI to treatment with Esprit BTK DRS ($n=173$) or PTA ($n=88$) between August 2020 and September 2022 at 50 global sites. Detailed baseline characteristics and primary safety and efficacy results have been previously published.¹³ As shown in Table 1, the mean age of the 261 enrolled patients was 72.6 years, with 32% identifying as women. The majority were White (59%), with smaller proportions of Asian (18%) and Black patients (12%). Medical histories were notable for hypertension in 93% of patients, dyslipidemia in 81%, and diabetes in 71%. RBC showed a near-even distribution, and half of the patients had wounds on the index limb at presentation.

Table 1. Key Baseline Patient Characteristics

Characteristic	Total (n=261)	DRS (n=173)	PTA (n=88)
Age, years	72.6±10.1	73.3±9.9	71.1±10.4
Women	83 (31.8)	56 (32.4)	27 (30.7)
Race and ethnicity			
White	154 (59.0)	98 (56.6)	56 (63.6)
Asian	47 (18.0)	36 (20.8)	11 (12.5)
Black	32 (12.3)	21 (12.1)	11 (12.5)
Other*	29 (11.1)	19 (11.0)	10 (11.4)
Hispanic or Latino	43 (16.5)	31 (17.9)	12 (13.6)
Medical history			
Tobacco use	138 (52.9)	91 (52.6)	47 (53.4)
Former smoker†	85 (32.6)	54 (31.2)	31 (35.2)
Never	123 (47.1)	82 (47.4)	41 (46.6)
Current	53 (20.3)	37 (21.4)	16 (18.2)
Hypertension	243 (93.1)	163 (94.2)	80 (90.9)
Dyslipidemia	212 (81.2)	140 (80.9)	72 (81.8)
Diabetes	184 (70.5)	123 (71.1)	61 (69.3)
Rutherford-Becker class‡			
4	135 (51.7)	90 (52.0)	45 (51.1)
5	126 (48.3)	83 (48.0)	43 (48.9)
Wound on index limb§	130 (49.8)	85 (49.1)	45 (51.1)
Toe	99 (53.8)	68 (59.1)	31 (44.9)
Calf	2 (1.1)	1 (0.9)	1 (1.4)
Foot	31 (16.8)	14 (12.2)	17 (24.6)
Heel	14 (7.6)	8 (7.0)	6 (8.7)
Shin	20 (10.9)	11 (9.6)	9 (13.0)
Sole	18 (9.8)	13 (11.3)	5 (7.2)
Ankle brachial index of target limb	0.88±0.32	0.87±0.32	0.91±0.33
Toe brachial index of target limb	0.49±0.29	0.51±0.31	0.46±0.24

Values are n (%) or mean±SD. DRS indicates drug-eluting resorbable scaffold; and PTA, percutaneous transluminal angioplasty.

*Native Hawaiian or Pacific Islander; American Indian or Alaska Native; Declined or unable to disclose.

†Former smoker: quit within the past month or >1 month ago.

‡Rutherford-Becker class 4 is ischemic rest pain, and class 5 is ischemic rest pain with minor tissue loss.

§Some patients presented with multiple wounds; therefore, the sum of wound locations exceeds the total number of patients.

||The number of patients for whom the ankle brachial index was available for analysis was 227 of 261 patients (87.0%), and the number of patients for whom a toe brachial index was available was 79 of 261 (30.3%). When the ankle brachial index was >1.4 or could not be measured reliably, a toe pressure measurement was obtained, and the toe brachial index was determined and used.

Baseline, Procedure, and Postprocedure Lesion Characteristics and DAPT Usage

Table 2 highlights key lesion and procedural characteristics. The mean reference vessel diameter was 2.9 mm, and the lesion length was 44 mm on average. Moderate calcification was observed in 28% of lesions, with severe calcification in 3% of lesions. Predilatation was

Table 2. Key Baseline, Procedure, and Postprocedure Lesion Characteristics

Characteristic	Total (n=261)	DRS (n=173)	PTA (n=88)	Difference* (95% CI)
Target lesions (site reported)				
Mean number of target lesions	1.0±0.2	1.0±0.2	1.0±0.2	0.0 (−0.0, 0.1)
Calcification				
Moderate	75 (27.7)	49 (27.4)	26 (28.3)	−0.9 (−12.5, 9.9)
Severe	8 (3.0)	6 (3.4)	2 (2.2)	1.2 (−4.5, 5.3)
Target lesion characteristics (core laboratory reported)				
Reference vessel diameter, mm	2.90±0.76	2.94±0.77	2.82±0.74	0.12 (−0.1, 0.3)
Lesion length, mm	44.1±30.9	43.8±31.8	44.8±29.1	−0.97 (−8.7, 6.8)
Target lesion location				
Anterior tibial	83 (31.8)	59 (34.3)	24 (27.0)	7.3 (−4.7, 18.3)
Posterior tibial	42 (16.1)	26 (15.1)	16 (18.0)	−2.9 (−13.2, 6.1)
Peroneal	49 (18.8)	28 (16.3)	21 (23.6)	−7.3 (−18.2, 2.5)
Tibioperoneal trunk	41 (15.7)	26 (15.1)	15 (16.9)	−1.7 (−11.9, 7.1)
Tibioperoneal trunk, posterior tibial	23 (8.8)	15 (8.7)	8 (9.0)	−0.3 (−8.7, 6.5)
Tibioperoneal trunk, peroneal	23 (8.8)	18 (10.5)	5 (5.6)	4.9 (−3.0, 11.2)
TASC II classification†				
A	130 (49.8)	83 (48.3)	47 (52.8)	−4.5 (−17.0, 8.1)
B	84 (32.2)	61 (35.5)	23 (25.8)	9.7 (−2.4, 20.5)
C	47 (18.0)	28 (16.3)	19 (21.3)	−5.0 (−15.8, 4.5)
DS preintervention, %	73.0±19.6	72.6±18.9	73.7±21.0	−1.1 (−6.3, 4.1)
Procedural characteristics (site reported)				
Predilatation performed	271 (100.0)	179 (100.0)	92 (100.0)	0.0 (−2.1, 4.0)
Postdilatation performed without complications	NA	176 (99.4)	NA	NA
Subject with bailout devices	5 (1.9)	0 (0.0)	5 (5.7)	−5.7 [−12.6, −1.8]
Postprocedure measurements (core laboratory reported)				
Treatment length, mm‡	60.9±40.7 (236)	59.1±39.7 (156)	64.5±42.5 (80)	−5.33 [−16.6, 6.0]
Residual DS after predilatation, %	29.8±12.5	30.0±12.6	25.0±8.7	5.0 (−14.8, 24.8)
Postprocedure in-segment residual DS, %	19.0±10.3	17.0±9.3	22.8±11.2	−5.8 (−8.6, −3.0)
Final DS <30%	224 (88.2)	163 (95.9)	61 (72.6)	23.3 (13.9, 33.8)
Residual flow limiting dissection	0 (0.0)	0 (0.0)	0 (0.0)	0.0 (−4.4, 2.2)

Values are n (%) or mean±SD. DRS indicates drug-eluting resorbable scaffold; DS, diameter stenosis; NA, not applicable; PTA, percutaneous transluminal angioplasty; and TASC II, Trans-Atlantic Inter-Society Consensus II.

*Differences in percentages are shown in percentage points.

†The Trans-Atlantic Inter-Society Consensus II classification indicates the anatomical pattern of atherosclerotic disease severity. No patients had class D.

‡The length of the vessel segment in which the device was implanted (for the Esprit BTK arm) or balloon angioplasty applied (for the percutaneous transluminal angioplasty arm).

performed in all cases, and postdilatation without complications was achieved in 99% of DRS cases. Bailout stenting was required in 5 patients (6%) in the angioplasty group. Postprocedure measurements indicated residual DS of 17% in the DRS group versus 23% in the PTA group, with 96% of lesions in the DRS group versus only 73% in the PTA group achieving a final DS of less than 30%. At 2 years, more patients in the PTA arm remained on DAPT (76.7% versus 66.1%) despite longer DAPT being part of the treatment protocol in the DRS arm and only short-term DAPT recommended in the PTA arm.

Patient Enrollment and Follow-Up

Patient flow through 2 years is shown in [Figure S1](#). Of the 261 patients initially enrolled, 173 were assigned to DRS treatment and 88 to PTA. Although clinical data were available for 87.9% of patients treated with DRS and 83% of those treated with PTA by the end of the 2-year follow-up period, some patients had to be excluded from the analysis because they did not have DUS or angiography data to evaluate patency at 2 years. As a result, the analysis included data for 122 patients in the DRS arm and 64 patients in the PTA arm.

Clinical Efficacy and Safety Outcomes

Clinical efficacy and safety outcomes at 2 years are summarized in Table 3.

The primary efficacy end point, a composite of primary patency and limb salvage, was observed in 61.5% of the DRS group compared with 32.8% of the PTA group at 2 years. This marked an absolute risk difference of 28.7% and an HR of 0.48 ($P=0.0004$; Figure 1). The individual components of the primary efficacy end point demonstrated similar trends, with the DRS group showing higher freedom from binary restenosis (64.8% versus 42.2%, absolute risk difference 22.6%) and clinically CD-TLR (88.5% versus 76.6%, absolute risk difference 12.0%). Freedom from above-ankle amputation remained high in both groups (93.4% for DRS versus 96.9% for PTA), whereas freedom from 100% total occlusion was slightly higher in the DRS group (81.1% versus 76.6%).

The primary safety end point, freedom from major adverse limb events at 2 years and perioperative death at

30 days, was assessed within the as-treated population and observed in 90.4% of patients in the DRS group compared with 95.9% in the PTA group. The KM curve for the primary safety end point is shown in Figure 2. At 2 years, the primary safety end point was observed in 91.6% of patients in the DRS group compared with 95.6% in the PTA group with an HR of 2.39 (95% CI, 0.68–8.38; $P=0.16$).

Binary restenosis of the target lesion, the first powered secondary end point, was identified in 80 subjects at 2 years. Of these, 35 cases were adjudicated by the CEC, with all but 1 also supported by imaging data (angiography or DUS). The remaining 45 cases were adjudicated solely by the angiographic or DUS core laboratories. In total, 30 cases were based on angiographic assessment and 49 on DUS assessment, and 1 was adjudicated by the CEC without available imaging. Binary restenosis was observed in 35% of patients in the DRS group and 58% in the PTA group among those who underwent imaging at the 2-year follow-up, yielding an absolute difference of –22.6% (95% CI, –36.47% to –7.52%), favoring DRS.

Table 3. Clinical Efficacy and Safety Outcomes

	Total	DRS	PTA	Difference [95% CI]*
Primary efficacy end point†‡ at 2 y (intent-to-treat population)	n=261	n=173	n=88	...
Composite of limb salvage and primary patency at 2 y	51.6% (96/186)	61.5% (75/122)	32.8% (21/64)	28.7% [13.6%, 41.8%]
Freedom from above-ankle amputation in index limb	94.6% (176/186)	93.4% (114/122)	96.9% (62/64)	–3.4% [–9.7%, 4.8%]
Freedom from 100% total occlusion of target vessel	79.6% (148/186)	81.1% (99/122)	76.6% (49/64)	4.6% [–7.1%, 17.7%]
Freedom from binary restenosis	57.0% (106/186)	64.8% (79/122)	42.2% (27/64)	22.6% [7.5%, 36.4%]
Freedom from CD-TLR	84.4% (157/186)	88.5% (108/122)	76.6% (49/64)	12.0% [0.9%, 24.5%]
Primary safety end point§ at 2 y (as-treated population)	n=260	n=170	n=90	...
Freedom from major adverse limb event at 2 y plus perioperative death (MALE + POD)	92.3% (193/209)	90.4% (123/136)	95.9% (70/73)	–5.5% [–12.1%, 2.8%]
Freedom from MALE at 2 y	93.3% (195/209)	91.9% (125/136)	95.9% (70/73)	–4.0% [–10.4%, 4.1%]
Freedom from above-ankle amputation in index limb	95.2% (199/209)	94.1% (128/136)	97.3% (71/73)	–3.1% [–8.8%, 4.2%]
Freedom from major reintervention on index limb	98.1% (205/209)	97.8% (133/136)	98.6% (72/73)	–0.8% [–5.1%, 5.3%]
Freedom from POD at 30 days	99.0% (207/209)	98.5% (134/136)	100.0% (73/73)	–1.5% [–5.2%, 3.6%]
First secondary end point¶ at 2 y (intent-to-treat population)	n=261	n=173	n=88	...
Binary restenosis of the target lesion at 2 y	43.0% (80/186)	35.2% (43/122)	57.8% (37/64)	–22.6% [–36.4%, –7.5%]
Second secondary end point¶ at 2 y (intent-to-treat population)	n=261	n=173	n=88	...
Freedom from above-ankle amputation in index limb, 100% total occlusion of target vessel, and CD-TLR at 2 y	70.4% (131/186)	75.4% (92/122)	60.9% (39/64)	14.5% [0.7%, 28.5%]

The denominators of the rates exclude subjects who left the study before the lower limit (702 days) of the 2-year follow-up window without any components of the primary end point. For the components of above-ankle amputation in index limb, major reintervention on index limb, and clinically driven target lesion revascularization, the data are based on clinical events committee adjudication through 730 days after the index procedure. Data for the components of 100% total occlusion of target vessel and binary restenosis of target lesion are based on clinical events committee adjudication or per imaging (core laboratory unscheduled angiography or duplex ultrasound) data, through 730 days if determined per event date, through 788 days if per angiography data, or 818 days if per duplex ultrasound data. CD-TLR indicates clinically driven target lesion revascularization; DRS, drug-eluting resorbable scaffold; MALE, major adverse limb event; POD, perioperative death; and PTA, percutaneous transluminal angioplasty.

*By Newcombe score method.

†Subjects were included in the analysis if the subjects had qualified 2-year core laboratory angiography data (or qualified core laboratory DUS data if the Angiographic were unavailable) or subjects who had the primary efficacy end point event.

‡Data are presented as the event-free rates (binary) through 2 years after the index procedure.

§Data for perioperative death are based on clinical events committee adjudication through 30 days after the index procedure. The major reintervention on index limb includes the creation of a new bypass graft, bypass graft revision, and the use of thrombectomy or thrombolysis per clinical events committee adjudication.

¶Data are presented as the event rates (binary) through 2 years after the index procedure.

‡Secondary end points were statistically powered at the 1-year time point only.¹⁴ The 2-year outcomes presented are descriptive and were not powered for formal hypothesis testing.

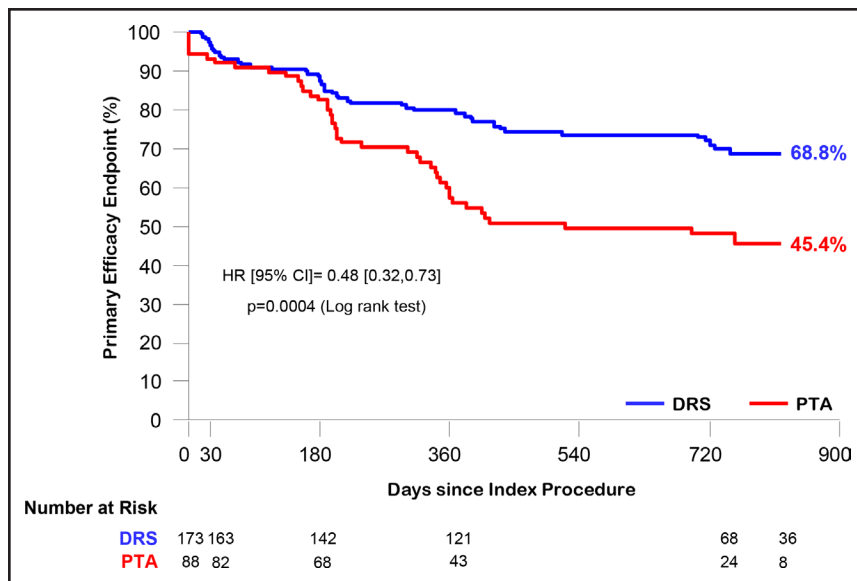


Figure 1. Primary efficacy end point by treatment through 2 years.

The primary efficacy end point was a composite of primary patency and limb salvage, defined as freedom from target limb amputation above the ankle, total occlusion of the target vessel, clinically driven target lesion revascularization, or binary restenosis of the target lesion.

KM estimates of 2-year binary restenosis, accounting for censoring and variable follow-up durations, were 28.5% in the DRS group and 48.2% in the PTA group (Figure 3). The hazard of restenosis was 85% higher in the PTA group (HR, 1.85 [95% CI, 1.19–2.88]; $P=0.005$, log-rank test). The separation of the KM curves became evident early and was maintained throughout follow-up.

A similar pattern was observed for CD-TLR at 2 years (Figure 4), with a significantly higher rate of revascularization in the PTA group compared with the DRS group (18.6% versus 9.7%; HR, 2.15 [95% CI, 1.04–4.46]; $P=0.034$, log-rank test).

Predictor Analysis

Predictors of the Primary Efficacy End Point

Univariate logistic regression analysis identified significant predictors of the primary efficacy end point at 2

years (Table S1). Treatment with a DRS was associated with lower odds of end point failure (odds ratio [OR], 0.31 [95% CI, 0.16–0.58]; $P=0.0003$). Other predictors included the presence and number of wounds on the target limb at index, longest lesion length, preintervention percent DS, postpredilatation residual percent DS, total occlusion at the treatment site, infrapopliteal grade, Global Limb Anatomic Staging System stage, and RBC. Multivariable regression (Table S2) confirmed these findings, with longer lesion, DRS treatment, RBC 5, and total occlusion at the treatment site as independent predictors.

Predictors of CD-TLR

The univariate analysis (Table S3) identified several factors associated with CD-TLR at 2 years in the intent-to-treat population. Treatment with a DRS was associated with 54% lower odds of CD-TLR compared with PTA (OR, 0.46; $P=0.054$). Other variables showing potential

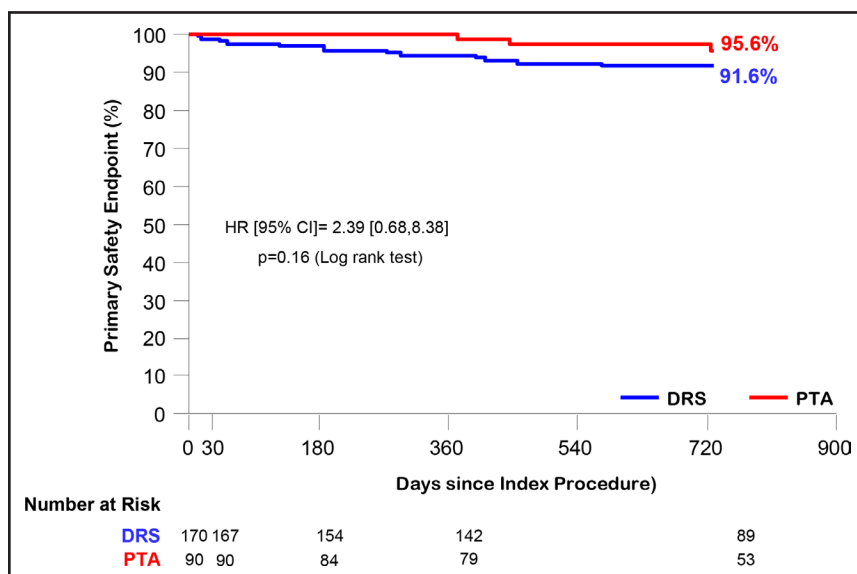


Figure 2. Primary safety end point (freedom from major adverse limb events and perioperative death).

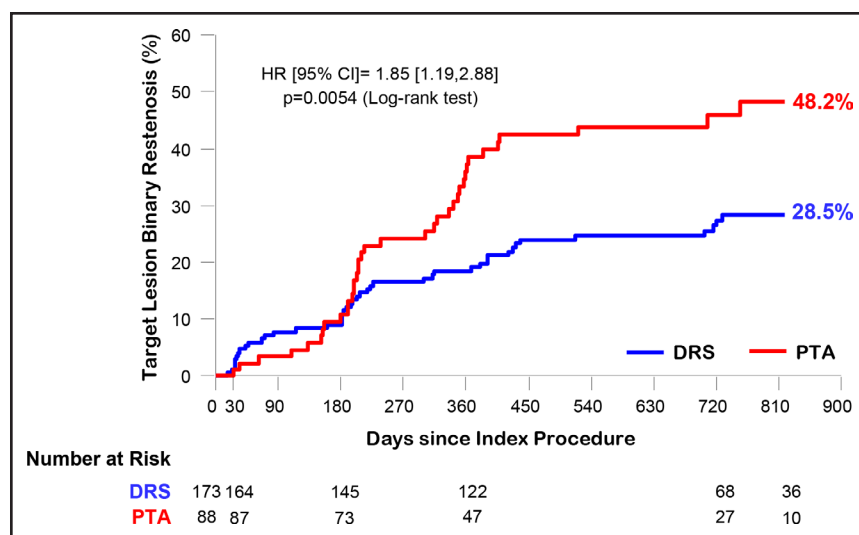


Figure 3. Binary restenosis (first secondary end point) of target lesion at 2 years.

associations with increased CD-TLR risk included White race (OR, 2.08; $P=0.089$) and diabetes (OR, 2.48; $P=0.078$). Significant predictors included previous amputation (OR, 4.77; $P=0.003$), number of wounds on the target limb at baseline (OR, 1.71; $P=0.008$), longest lesion length (OR, 1.02; $P=0.009$), preintervention percent DS (OR, 1.06; $P=0.008$), residual percent DS after predilatation (OR, 1.04; $P=0.011$), Global Limb Anatomic Staging System stage ≥ 2 (OR, 3.02; $P=0.032$), and RBC 5 (OR, 2.81; $P=0.014$).

In the multivariate model (Table S4), previous amputation on the target limb remained the strongest independent predictor of CD-TLR (OR, 5.21; $P=0.005$). Preintervention percent DS (OR, 1.06; $P=0.008$) and number of wounds on the target limb at baseline (OR, 1.55; $P=0.044$) also remained significant. Treatment with DRS was associated with a 57% reduction in the odds of CD-TLR compared with PTA (OR, 0.43; $P=0.062$). Although this did not reach statistical significance, the consistent direction and magnitude of effect

across both univariate and multivariate models may suggest a clinically meaningful benefit, warranting further studies.

Subgroup Analysis

Primary Efficacy End point.

The forest plot analysis (Figure S2) demonstrated that DRS consistently reduced the risk of adverse events compared with PTA across most subgroups, including patients with and without diabetes, varying levels of calcification, presence of inflow or nontarget lesions, and across lesion length terciles. No statistically significant interactions were observed (all P for interaction >0.1), indicating a consistent treatment effect. Notably, in patients with mixed-etiology wounds, the event rate was numerically higher with DRS (73.7% versus 66.7%; relative risk, 1.11 [95% CI, 0.59–2.07]), though this difference was not statistically significant ($P=0.388$).

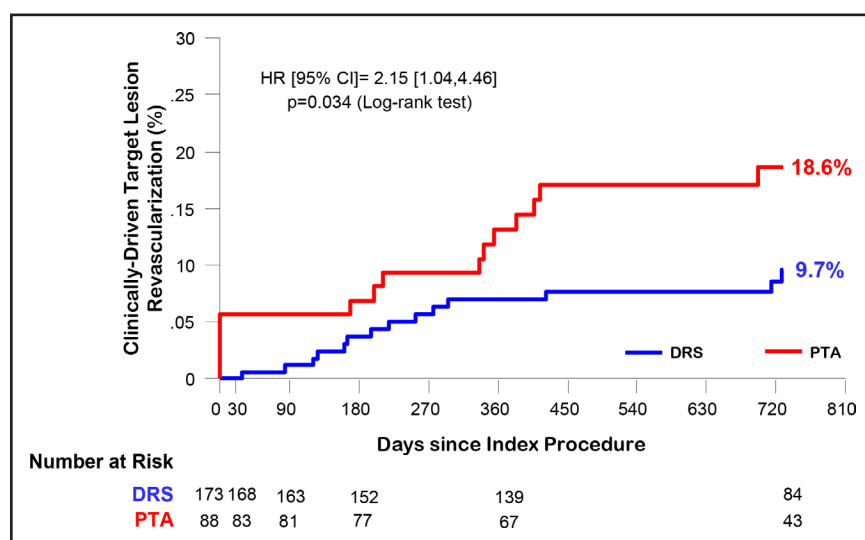


Figure 4. Clinically driven target lesion revascularization by treatment at 2 years.

Clinically Driven Target Lesion Revascularization

Subgroup analysis of CD-TLR at 2 years is presented in Figure S3. DRS generally demonstrated lower or comparable CD-TLR rates across most subgroups, including patients with and without diabetes, different calcification severities, and varying lesion lengths. A significant interaction was observed based on inflow lesion treatment ($P=0.037$), with DRS reducing revascularization risk only when no inflow lesion was treated (6% versus 25%; RR, 0.24 [95% CI, 0.09–0.64]). Additionally, in patients with mixed-etiology wounds, DRS was associated with a markedly lower CD-TLR rate (4.8% versus 37.5%; RR, 0.13 [95% CI, 0.02–1.05]), although wide CIs limit interpretation. No other significant interactions were observed.

DISCUSSION

Atherosclerotic disease in patients with CLTI is often multilevel with a high prevalence of infrapopliteal involvement.⁹ These lesions are particularly challenging to treat because of dense calcification, small vessel diameters encountered below the knee, the high likelihood of long-segment occlusions, and a high risk of dissection, recoil, and restenosis in these arteries following intervention.^{16,17} Multiple clinical trials have been undertaken in attempts to develop new therapies for infrapopliteal arteries that solve some of these critical challenges, including ones evaluating drug-eluting metallic stents and paclitaxel-based drug-coated balloons, but each failed to meet the primary end point. Consequently, before the LIFE-BTK trial and approval of the Esprit BTK DRS, PTA remained the default treatment strategy for tibial arteries.^{18,19} DRS devices have the benefit of scaffolding to support the vessel against recoil and dissection, limus-based drug elution to prevent intimal hyperplasia and biologic restenosis, and the ability to be absorbed when no longer needed in the vessel, allowing for the potential of leaving a healthy native vessel with return of vasomotion and freedom from permanent implants that could affect future need for distal intervention or surgical bypass options.^{12,20} The performance of this first-of-its kind device in the tibial vasculature was demonstrated in a previous publication reporting the 12-month results of the LIFE-BTK multicenter randomized controlled clinical trial, in which the primary efficacy end point (ie, no events occurred) was achieved in 74% of patients in the DRS group compared with 44% in the PTA group, representing a highly statistically significant difference of 30 percentage points.¹³

Although these results of the Esprit BTK DRS at 12 months were favorable, it is important to note that recidivism and recurrent wounds are not uncommon in patients with CLTI, and wounds with a large volume of tissue loss will sometimes require many months for successful healing.² Furthermore, in CLTI patients treated for ischemic rest pain, loss of patency is often associated with recurrent

symptoms. For these reasons, successful treatment of CLTI patients not only requires short-term success but should also be accompanied by sustained patency to maintain limb salvage and avoid the need for reintervention.

This report demonstrates that the Esprit BTK DRS may play an important role in achieving these goals, as the device maintained superior patency results compared with PTA out to 2 years with equivalent safety profiles. Equally importantly, there was little loss of patency between the 1-year and 2-year time points in the DRS arm (with only 5.4% difference by KM analysis), suggesting that the antirestenotic properties of the device may help achieve the long-term patency these patients need for optimal outcomes. Finally, not only did the DRS significantly improve primary patency rates, but it also demonstrated a markedly lower incidence of binary restenosis and CD-TLR compared with PTA, further supporting its potential to reduce the need for repeat revascularization procedures. Interestingly, the KM curve for binary restenosis (Figure 3) shows a slightly higher rate of early events in the DRS group during the first 180 days, followed by a clear and sustained benefit thereafter. This pattern may reflect the dual-phase mechanism of the scaffold; early restenosis events could be influenced by lesion complexity, procedural variability, or patient-specific biological factors, whereas the later divergence likely reflects the delayed antiproliferative effect of everolimus. These findings suggest that the long-term efficacy of the scaffold may be primarily pharmacological, with mechanical benefits potentially influenced by procedural variability in the early postprocedural phase. Further investigation of the temporal dynamics of restenosis in scaffold-treated lesions may help optimize patient selection and procedural techniques. Overall, these results underscore the robust support mechanism of the scaffold, which, combined with its drug-eluting properties, provides a comprehensive solution for managing restenosis in BTK interventions. These results highlight the unique ability of the DRS to address the limitations of previous treatments. The efficacy of the DRS in maintaining long-term arterial patency and reducing the risk of restenosis underscores its promise as a next-generation solution for complex infrapopliteal disease.

The analysis of efficacy and CD-TLR predictors provided valuable findings on factors influencing treatment outcomes with the DRS. DRS treatment was identified as an independent predictor of improved primary efficacy, significantly impacting long-term arterial patency and limb salvage. On the other hand, longer lesion length, RBC 5, and total occlusion at the treatment site were associated with higher event rates, indicating that more complex lesions are linked to worse outcomes in both arms. Additionally, CD-TLR risk was highest in patients with previous amputation on the target limb and those with extensive baseline disease burden, including greater preintervention stenosis and wound burden. Notably, DRS use was

associated with a trend toward reduced CD-TLR risk, suggesting that DRS treatment may offer substantial benefits in reducing the need for repeat revascularization in high-risk patients. This is particularly important in the context of CLTI, in which the risk of major amputation is high and the need for long-term, durable solutions is crucial.²¹

Subgroup analyses provided additional insights into the performance of the DRS across different patient populations. The scaffold treatment consistently reduced the risk of primary efficacy events compared with angioplasty, with the strongest effect seen in nondiabetic patients, those with RBC 4 disease, and patients with shorter lesions or no inflow lesion treatment. These findings suggest that the scaffold may offer particularly strong benefits in these subgroups, further emphasizing its potential for broader applicability in CLTI treatment. DRS significantly reduced events across patients with or without nontarget lesion treatment and in those with no or mild calcification. Although the benefit was less pronounced in more heavily calcified lesions, outcomes remained favorable. In patients with mixed-etiology wounds, the scaffold was associated with a slightly higher event rate than PTA but with no significant interaction, indicating that further exploration is needed in this subgroup. For CD-TLR at 2 years, the DRS reduced revascularization rates, particularly in patients without inflow lesion treatment. A significant interaction indicated that the DRS had a differential treatment effect in these patients, suggesting that those without inflow lesions may benefit the most from DRS treatment. Overall, the DRS demonstrated a consistent reduction in CD-TLR rates regardless of other factors, such as diabetes status, RBC, or lesion calcification severity. This further reinforces the robust efficacy of the DRS across diverse patient subgroups. Whereas this trial compared scaffold-based therapy with PTA alone, future studies evaluating the Esprit BTK scaffold against drug-eluting stents or provisional stenting strategies will be important to further define its role in the broader BTK treatment landscape.

This trial had several limitations that should be considered when interpreting the data. First, after an interim analysis (with investigators blinded to treatment arms) during the ongoing enrollment phase, binary restenosis was added to the primary efficacy end point to better capture clinically relevant hemodynamic outcomes. To assess the impact of this change, the original primary efficacy end point was retained as a powered secondary end point. Second, the trial population predominantly included patients with noncomplex, short, and mildly to moderately calcified lesions (approximately 80% Trans-Atlantic Inter-Society Consensus II A/B), which may not fully reflect the broader CLTI population encountered in routine clinical practice, in which long, heavily calcified BTK lesions and Trans-Atlantic Inter-Society Consensus II C/D classifications are more common. Therefore, the generalizability of these findings to more complex lesion

subsets remains limited. Ongoing studies, such as the Esprit BTK postapproval study, are expected to provide further insights into the performance of the scaffold in more complex, real-world anatomies. Third, mandatory predilatation in the scaffold group may have influenced the results to some extent; however, this technique is recognized as a crucial factor in achieving optimal outcomes with these scaffolds. Fourth, the close supervision in the trial likely reduced the expected incidences of amputation and CD-TLR compared with real-world clinical practice. Fifth, the use of DRS was limited to the proximal two-thirds of infrapopliteal arteries, which may limit the applicability of these findings to other locations.

Conclusions

The 2-year results from the LIFE-BTK trial demonstrate that the Esprit BTK everolimus-eluting resorbable scaffold significantly improves primary patency compared with balloon angioplasty in patients with CLTI. The scaffold also reduces rates of binary restenosis and CD-TLR while maintaining a comparable safety profile. These findings support the Esprit BTK scaffold as a promising treatment option for infrapopliteal artery disease in appropriately selected patients with CLTI.

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Supplemental Material

Checklist

Figures S1–S3

Tables S1–S4

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